Dikshant Agrawal

M.Sc. Embedded Systems and Microelectronics (EE & IT)
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Portfolio | LinkedIn | GitHub

Profile

Embedded Systems Master's student with 3 years of industry experience writing embedded software components in C/C++, software integration, testing, debugging, and validation. Seeking a mandatory internship in the embedded software development.

Work Experience

Research Student – Firmware Development & Test Mar. 2026 – Present
Hochschule Darmstadt University of Applied Sciences *Darmstadt, Germany*

- Developing embedded software drivers in C on a PSoC 5LP-based wireless remote of student cars: Zigbee CDD/RF stack extended from 1:1 to a 1:n fleet, backward-compatible with deployed vehicles.
- Software integration, testing and debugging with the car team via a shared Gitlab CI/CD workflow; write low-level C drivers and unit tests.

Embedded Systems Engineer Jan. 2023 – Aug. 2025
IITI DRISHTI CPS Foundation – Translational Research Park, IIT Indore *Indore, India*

- Built a real-time body vital-monitoring firmware in C; validated at three clinical sites.
- Developed Python tooling for sensor calibration, signal validation, reducing manual effort by 70%.

Research and Development Engineer Sep. 2022 – Jan. 2023
Scientech Technologies Pvt. Ltd. *Indore, India*

- Designed wireless embedded training boards for energy monitoring; adopted by 50+ colleges.

Hardware Design Intern Jan. 2022 – Jul. 2022
Einfochips Pvt. Ltd. – an Arrow Company *Ahmedabad, India*

- Verified I2C, SPI, UART during board bring-up; JTAG-based firmware debugging and fault detection.

Education

Hochschule Darmstadt University of Applied Sciences Sep. 2025 – Present
M.Sc. in Electrical Engineering & IT (Embedded Systems & Microelectronics) *Darmstadt, Germany*

- Coursework: Embedded Architecture & OS, Safety in Embedded Control Systems - ISO26262, MISRA C.

Shri Vaishnav Inst. of Technology and Science (SVVV) Jul. 2018 – Aug. 2022
B.Tech. in Mechatronics *Indore, India*

- Thesis: IoT-based baby monitor on ESP32 with real-time sensor analysis and alert system.

Projects

Electronic Gas Pedal Apr. 2026 – Jun. 2026

- Closed-loop PID engine control on an AUTOSAR-style RTE with Erika (OSEK); analysed in MATLAB.

UART-CAN Gateway in C++ Oct. 2025 – Dec. 2025

- Ring-buffers and layered port hierarchy routing events between UART and CAN.

More projects: dikshant-agrawal.github.io

Skills

Programming	C, C++, Python, MATLAB
Embedded	AUTOSAR-style CDD/RTE, Erika OSEK/FreeRTOS, low-level drivers, Cortex-M
Protocols	CAN, UART, SPI, I2C, Zigbee, MQTT
Tools & Test	Git, GitLab, CI/CD, unit testing, JTAG/SWD, logic analyzer, Calude Code.
Languages	English: Fluent, German: A2-B1, Hindi: Native

Darmstadt, July 11, 2026
Dikshant Agrawal